

Will You be Approved?

Predicting Credit Card Approval

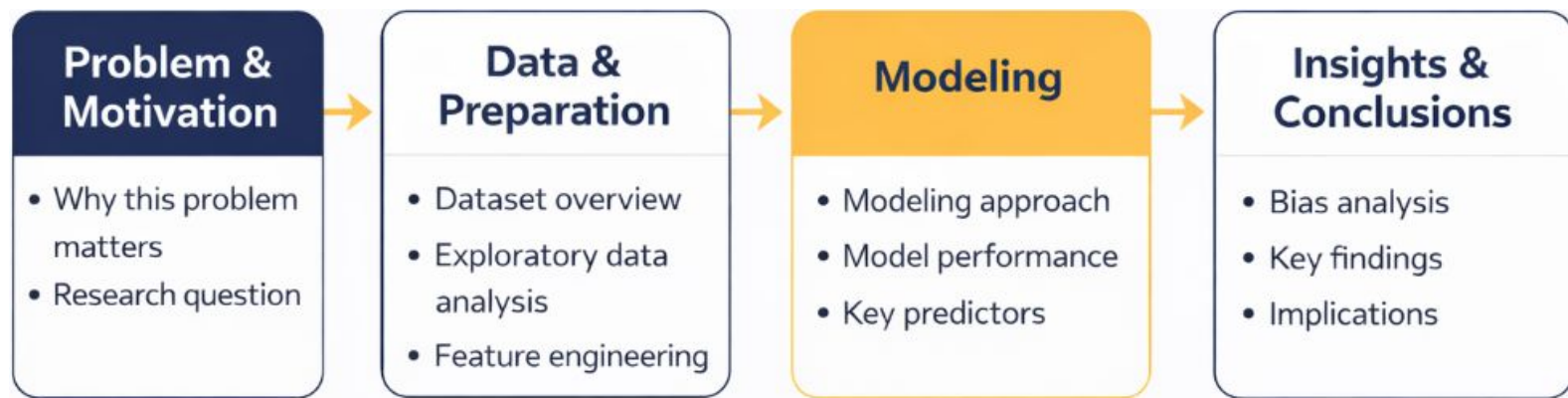
Predictive Analytics - Group 2



Jonathan, Edward, Tyler,
Justin, Ann



Presentation Overview



Why This Problem Matters



Context:

Credit approval determines financial access

Challenge:

Banks must balance risk and approval rates

Opportunity:

Predictive analytics enables data-driven lending



Research Question & Hypothesis

Can applicant financial characteristics predict credit card approval decisions?

Status Quo

Credit approval decisions traditionally rely on manual underwriting or rule-based evaluation

Manual reviews → slower decision workflows

Rule-based criteria → limited predictive power

Hidden patterns in data → overlooked by traditional methods



Research Hypothesis

Applicant historical financial characteristics significantly improve Bank's abilities to identify customers who would be approved

Possible Key Predictors:

Credit Score

Income

Debt Levels



Dataset Overview

690 observations (applications)

15 variables

0 missing values

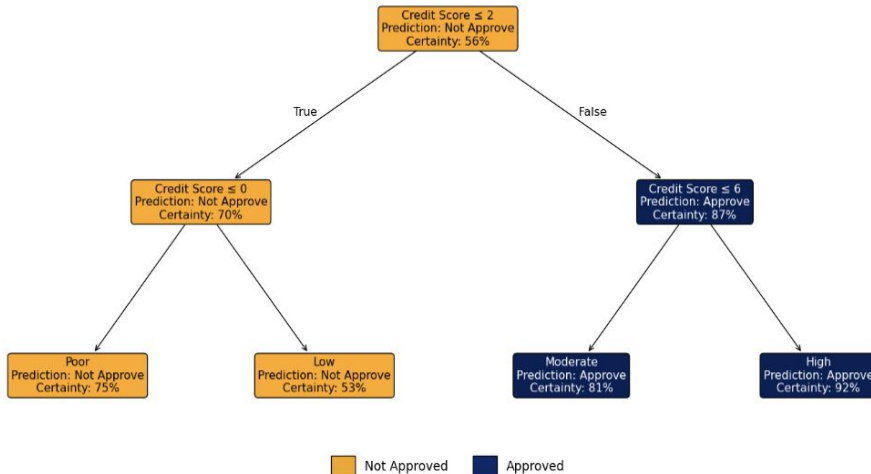
Financial Indicators	Employment Indicators	Applicant Information	Bank Relationship	Other	Target Variable
Credit Score	Employed	Age	Bank Customer	Zip Code	Approved (Yes/No)
Income	Years Employed	Gender		Drivers License	
Debt	Industry	Marital Status		Citizenship	
Prior Default		Ethnicity			

Data Transformation



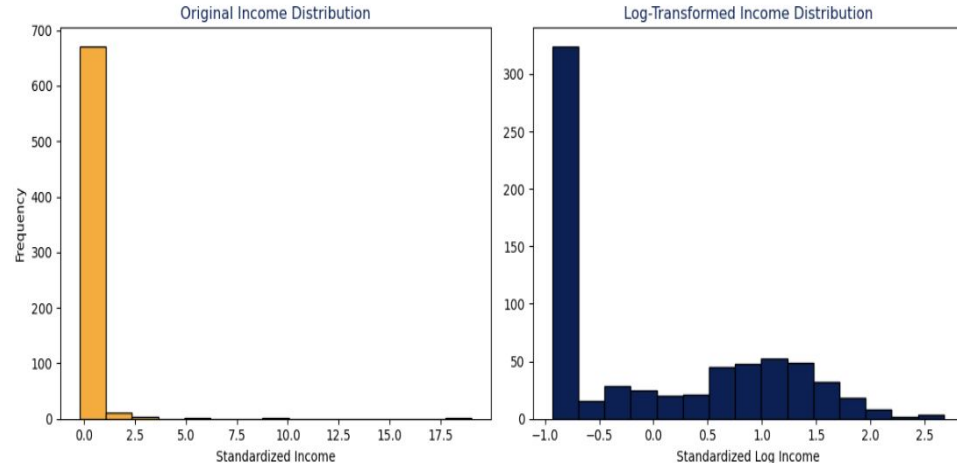
Binning

Scaled CreditScore using Decision Tree binning.



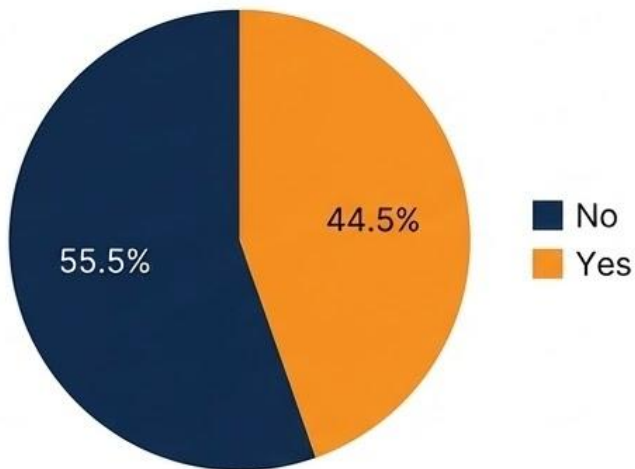
Log Transformation

Income and Debt variables showed right skewness, apply log transformations.



Key Insights from EDA

Approval Distribution



Target Variable is evenly distributed



Numeric Strong Predictors

- CreditScore
- YearsEmployed



Categorical Strong Predictors

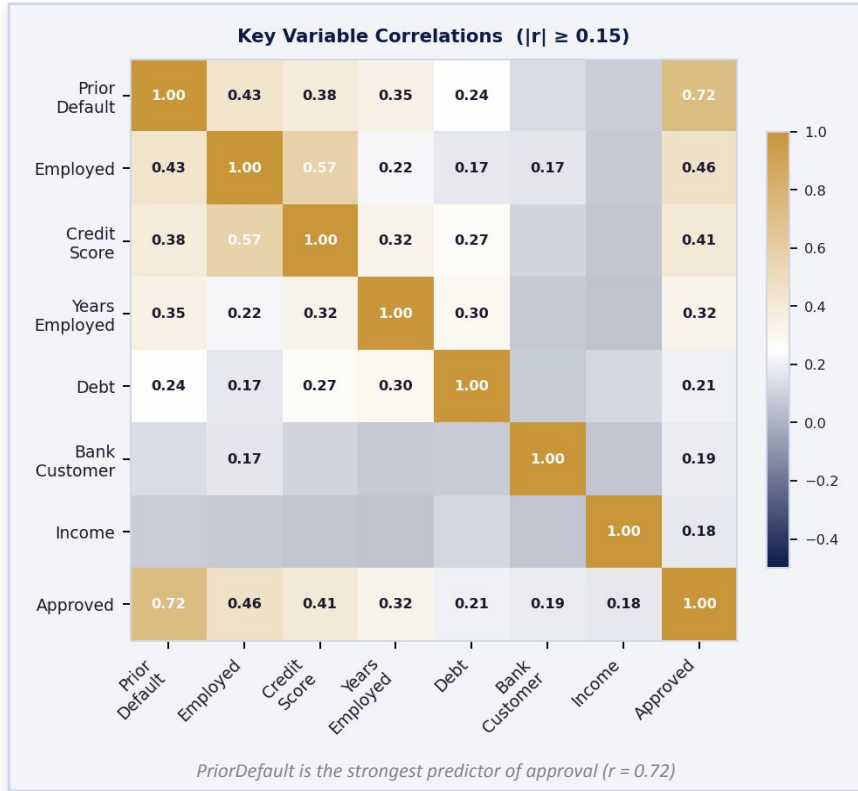
- PriorDefault
- BankCustomer
- EmploymentStatus
- MaritalStatus



Cautions

- Ethnicity
- Debt
- CitizenshipStatus
- Income

Data Preparation



Feature Selection

PriorDefault, Employed, CreditScore & YearsEmployed selected as strongest predictors of approval.

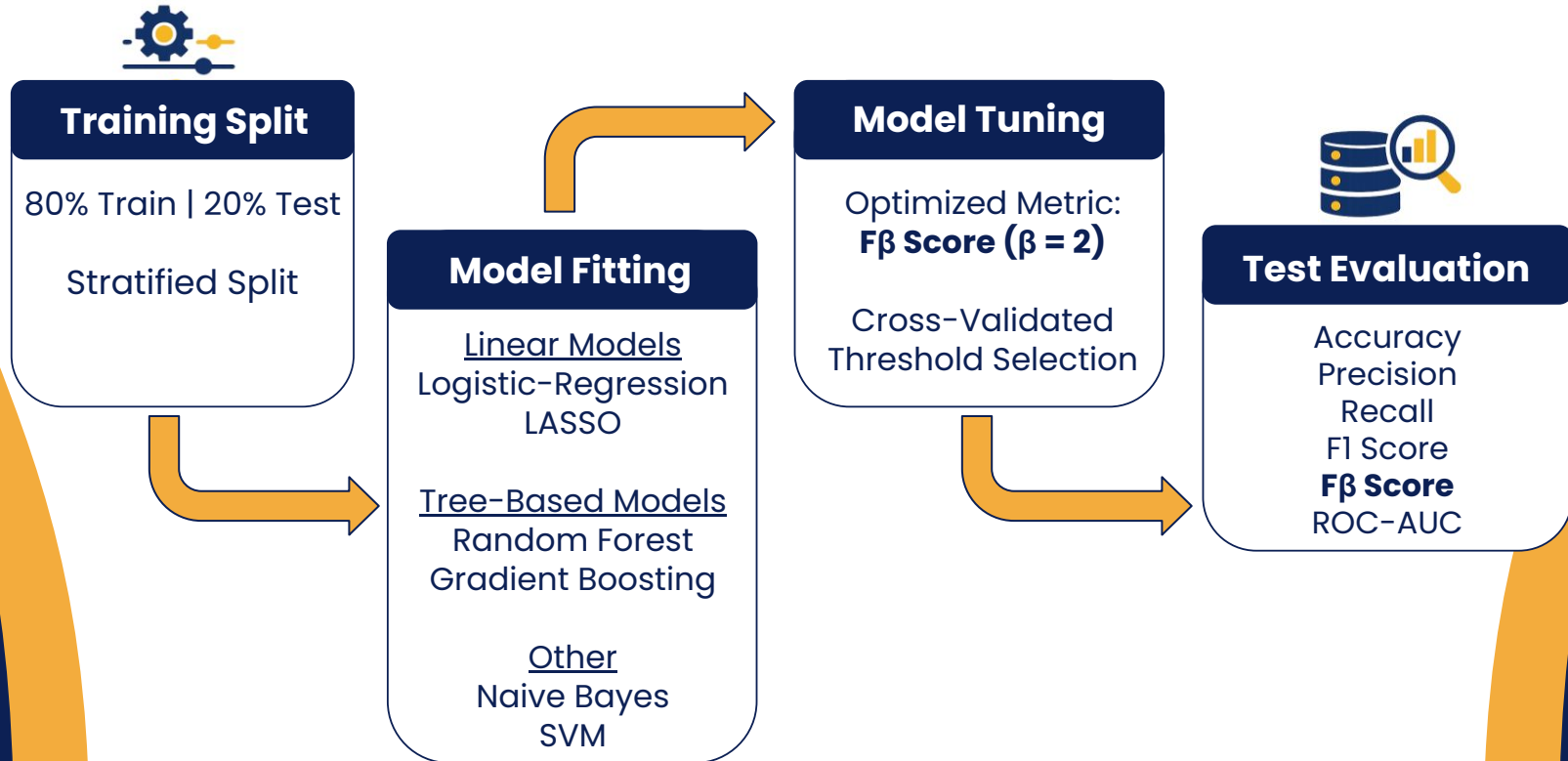
Removals

Age, MaritalStatus, CitizenshipStatus, Gender & DriversLicense removed due to multicollinearity or low predictive value.

Final Dataset

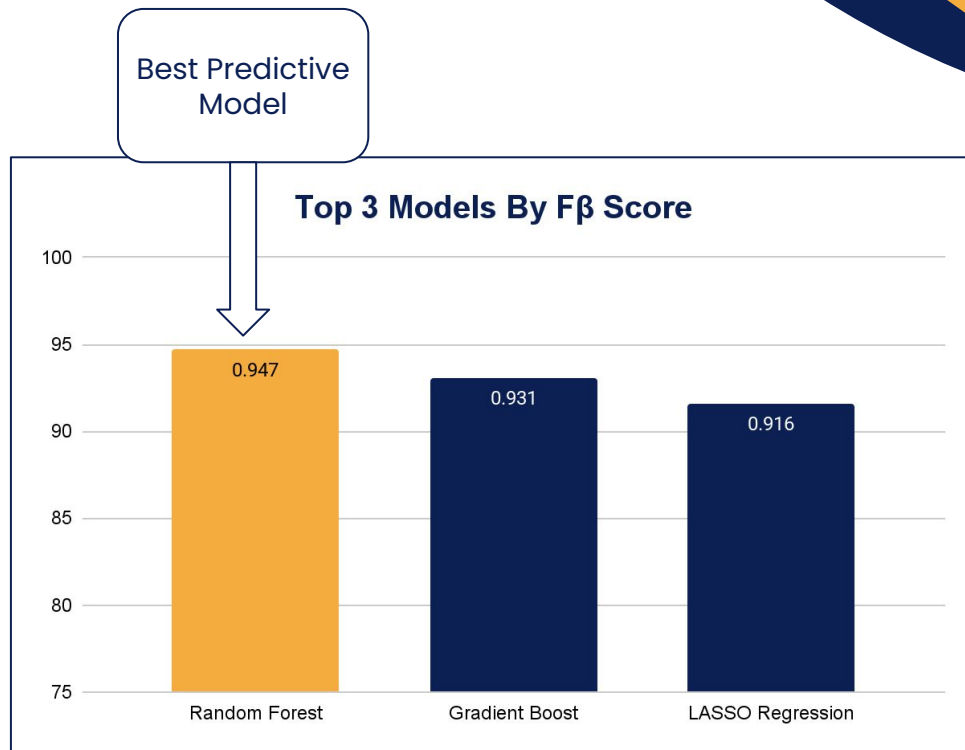
Remaining variables used to train and evaluate all prediction models.

Modeling Approach



Model Performance

Model	Recall	Test F β
Random Forest	1.00	0.95
Gradient Boost	0.97	0.93
LASSO	0.93	0.92
Elastic Net	0.93	0.92
SVM	0.92	0.90



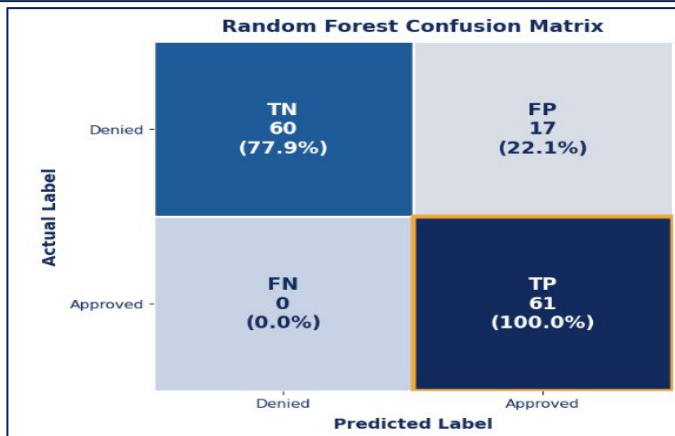
Final Chosen Models

Best Predictive Model Random Forest

Highest $F\beta$ (**0.947**)

Perfect Recall ~ **captured all likely approved applicants** in the test set

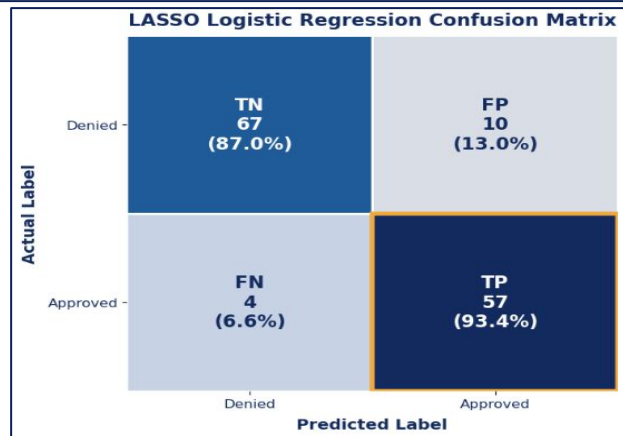
Captures **nonlinear interactions between predictors**



Most Interpretable Model LASSO Regression

Strong performance with **clear coefficient interpretability**

Allows for better understanding of primary drivers of approval



Key Model Insights

Positive Drivers

High Credit Score

Higher Income

Employment Status

Years Employed

Existing Bank Customer

Debt x Moderate Credit
Interaction

Negative Drivers

Prior Default History

Strongest Negative Predictor

Healthcare Industry

Consumer
Discretionary Industry

Insights

Financial Stability
→ Drives Approvals

Nonlinear Relationships
Captured by Ensemble
methods

Certain demographic and industry variables, including Latinos as well as Healthcare Consumer Discretionary workers, showed negative coefficients. These relationships are explored further in the next analysis

Bias Analysis

Approval Rate by Ethnicity

Group	Approval	Yrs Employed	Income (Log)	Prior Default	Bank Customer	Debt
Latino	14%	1.41	2.58	.77	.65	5.38
Non-Latino	42–63%	2.30	3.01	.45	.77	4.70

Latino applicants approved at 3–4× lower rates despite similar debt profiles

Approval Rate by Industry

Industry	Approval	Yrs Employed	Income (Log)	Prior Default	Debt
Utilities	~84%	3.10	3.92	.16	6.10
Energy	~45%	2.20	3.11	.16	4.20
Healthcare	~13%	1.32	2.89	.08	5.84

Healthcare applicants face approval rates comparable to Latino applicants, despite lowest default risk

Key Takeaway

Approval rates vary dramatically across both ethnicity and industry, independent of financial risk indicators like prior default and debt.

Latino applicants are approved at just 14% vs. 42–63% for non-Latino applicants.

Healthcare workers face a roughly 13% approval despite the lowest default rates in the dataset.

Conclusion

Predictive Performance

Random Forest achieved the strongest predictive performance, with the highest $F\beta$ score and perfect recall, making it the best model for minimizing missed approvals.

Interpretability

LASSO Logistic Regression provided transparent, actionable insights into the key drivers of approval decisions. This enables compliance-friendly explanations.

Strategic Impact

Combining predictive accuracy with interpretability enables more effective credit screening, improved approval targeting, and better risk management.

Together, these models provide a robust, explainable credit approval framework between performance, fairness, and regulatory transparency.

Data Preparation

Feature Selection

Prior Default History, Employment Status, CreditScore, & YearsEmployed are strongest predictor

Removals

Age, MaritalStatus, CitizenshipStatus, Gender & DriversLicense removed due to multicollinearity or simplification

Final Dataset

Variables used to train and evaluate prediction models

